

User defined functions in Mysql

When to write function

1. When you want to return single value
2. These functions we can call from select statement or where clause, or we can call then inside procedure or function

Syntax for writing function

```
CREATE FUNCTION func_name (parameters)
RETURNS return_type
BEGIN
    statements
END//
```

To see the list of all functions

show function status where db='test';

or

```
SELECT
    routine_schema AS "Database",
    routine_name,    ROUTINE_DEFINITION
FROM
    information_schema.ROUTINES
WHERE
    routine_type = 'FUNCTION'
ORDER BY
    routine_schema ASC, routine_name ASC;
```

In PLSQL, if you want to return single value as output, then use function

to write functions, it is necessary to assign following variable

set global log_bin_trust_function_creators=1;

1. write a function to generate email, by concatenating first 3 letters of name and last 3 letters of job and concatenate '@mycompany.com'.

```

crate procedure displayallemails(dno int)
begin
    declare vempno,vdeptno, vset int default 0;
    declare vjob,vename varchar(30)
    declare vem varchar(50);
    declare empcur cursor for select empno,ename,job,deptno from emp
                                where deptno= dno;
    declare continue handler for NOT FOUND set vset=1;
    open empcur;
    label1: loop
        fetch empcur into vempno,vename,vjob,vdeptno;
        if vset=1 then
            leave label1;
        end if;
        set vem=generateEmail(vename,vjob);
        select vempno,vename,vjob,vdeptno,vem;
    end loop
    close empcur;

end//

```

```

delimiter //
create function generateEmail(nm varchar(20),
    ejob varchar(20)) returns varchar(30)
begin
    declare vem varchar(30);
    set vem=concat(substr(nm,1,3),',',
        right(ejob,3),'@mycompany.com');
    return vem;
end//

to use function
select empno,ename,job, generateEmail(ename,job)

```

from emp;

2. calculate experience of a employee

delimiter //

create function calcexp(edate date) returns int

begin

declare vexp int;

set vexp=timestampdiff(year,edate,curdate());

return vexp;

end//

delimiter ;

3. write a function to calculate discount rate for room

```
delimiter //
create function calcDiscount(prloc varchar(20),prate double(9,2))
returns double(9,2)
begin
    declare vdiscount double(9,2) default 0;
    if prloc="seaview" then
        set vdiscount=prate*0.10;
    elseif prloc="hill side" then
        set vdiscount= prate*0.15;
    else
        set vdiscount=prate*0.08;
    end if;
    return vdiscount;
end//
delimiter ;
```

Exception Handling

Handling exceptions

There are 2 types of exception handlers_action

continue	resume the code after handling the exception
exit	stop the code after handling the exception

syntax

declare <handler_action> handler for <exception> statements

<exception>

1. SQLEXCEPTION
2. mysqlerror code
3. SQLSTATE value
4. NOT FOUND

```

create procedure insdeptrec(did int,dname varchar(20),dloc varchar(20))
delimiter //
create procedure insdeptrec(did int,pdname varchar(20),pdloc varchar(20))
begin
    declare exit handler for SQLEXCEPTION select "error occurred";
    insert into dept values(did,pdname,pdloc);
    select did,pdname,pdloc;
end//
delimiter ;

```

3. Handle check exception for expdate< mfgdate
 , also handle foreign key exception for wrong category value,
 And for primary key, if duplicate pid is entered

```

delimiter //
Create procedure insertproduct1(pid int,pnm varchar(20),qty int, pr
double,mdt date,edt date, catid int)
Begin
    Declare exit handler for SQLEXCEPTION select "Error occurred";
    Declare exit handler for 1062 select "duplicate product id entry";
    Declare exit handler for 1452
        begin
            select "wrong category value";
            Select "pls enter correct catagory value";
        End;
    Declare exit handler for 3819 select "incorrect expiry date" ;
    insert into product values(pid,pnm,qty,pr,mdt,edt,catid);
    Select concat(pid,"",pnm,"",qty,"",pr,"",mdt,"",edt,catid,"inserted");
End//
delimiter ;

```

4. Using user defined exception

1. Declare user defined exception
 declare category_exists condition for SQLSTATE '45000';

2. Declare handle for the user defined exception

```

declare exit handler for category_exists
begin
    select "Error:duplicate category name" errormsg;
end;

```

3. Based on condition raise the user defined exception, signal statement will raise exception explicitly

```

if exists (select * from category where cname=pcnam) then
    signal category_exists set MESSAGE_TEXT = "The given category name exists";

```

```

delimiter //
create procedure insertcategory(pcno int,pcnam varchar(20),pcdesc varchar(20))
begin
    declare category_exists condition for SQLSTATE '45000';

```

```

declare exit handler for category_exists
begin
    select "Error:duplicate category name" errmsg;
end;
if exists (select * from category where cname=pcnam) then
    signal category_exists set MESSAGE_TEXT="The given category name exists";
else
    insert into category values(pcno,pcnam,pcdesc);
    select "category record inserted";
end if;
end//
delimiter ;

```

to see the definition of the procedure

select ROUTINE_DEFINITION

-> from information_schema.Routines

-> where specific_name='getremark1';

to see the list of procedures in the database

show procedure status where db='iacsd0925'

to see the list of functions in the database

show function status where db='iacsd0925'

triggers

When to write the trigger

1. to monitor operation on a table
 - a. if you want to keep monitoring , how DML operations are happening on any table by which user or at what time
2. Data gathering for data analysis
 - a. If any product cost or qty is changing we may keep track of old values and new values, and use it for analysis purpose
3. To manage correctness of denormalize

example

if we have following tables

account(aid, custid,type, balance)

customer(cid, cname,mobile)

for every transaction if we need to use mobile for sending SMS, then we always need to take join of account and customer table

so to improve performance, we can avoid join by adding mobile number in account table, and make our data denormalized

account(aid, custid,type, balance,mobile)

customer(cid, cname,mobile)

so in this case to manage correctness of mobile number, we need trigger, which will update mobile in accounts table as soon as mobile changes in customer table

This is helpful to manage correctness of data.

4. to manage dml operation on some complex views, but the triggers on views are not available in mysql, but available in oracle.

It is task which gets executed automatically.

1. every trigger receives 2 special variables **old and new** in mysql and in oracle the name is :new and :old

	old	new
insert	null	new updated values
delete	old existing values from table	null
update	old existing values from table	new updated values in table

```
create table dept_check(
```

```
    olddno int,  
    newdno int,  
    dname varchar(20),  
    oldloc varchar(20),  
    newloc varchar(20),  
    uname varchar(20),  
    chdt datetime,  
    action varchar(20));
```

```
delimiter //
```

```
create trigger updatedept before update on dept  
for each row
```

```
begin
```

```
    insert into dept_check
```

```
    values(old.deptno,new.deptno,old.dname,old.loc,new.loc,user(),now(),'update');
```

```
end//
```

```
create table emp_check(
```

```
    eno int,  
    ename varchar(20),  
    oldsal float(9,2),  
    newsal float(9,2),  
    oldjob varchar(20),  
    newjob varchar(20),  
    uname varchar(20),  
    changedate datetime,  
    action varchar(20));
```

```
----create update trigger
```

```
create trigger updateemptr after update on emp  
for each row
```

```
    insert into emp_check
```

```
    values(old.empno,old.ename,old.sal,new.sal,old.job,new.job,user(),now(),'update');
```

-----to test whether trigger gets executed

```
update emp
set sal=sal*1.10,job='sr. clerk'
where job='clerk';
```

-----check old and new salary and job entries are there in the emp_check table

```
select * from emp_check;
```

```
create trigger insertemptr after insert on emp
for each row
insert into emp_check values(new.empno,new
.ename,null,new.sal,null,new.job,user(),now(),'insert');
```

```
create trigger deleteemptr after delete on emp
for each row
insert into emp_check values(old.empno,old
.ename,old.sal,null,old.job,null,user(),now(),'delete');
```

account

accid	custid	balance	type	mobile
100	1001	455666	saving	6666
101	1001	55555	current	5555
102	1001	55666	demat	5555
103	1002	4444444	saving	44444

customer

custid	cname	mobile
1001	Kishori	6666
1002	Rajan	44444
1003	Revati	555444

Using trigger for maintaining denormalized data

Update customer Set mobile=6666 Where custid=1001	Create trigger updateacc after update on customer For each row begin Update account Set mobile=new.mobile Where custid=old.custid; End//
--	--

Normalize the data

If the data is normalized then redundancy of data will be as less as possible

accid	custid	cname	mobile	balance	type	RM no	RM name
100	1001	Kishori	55555	455666	saving	1	Sahil
101	1001	Kishori	33333	55555	current	1	Sahil
102	1001	Kishori	33333	55666	demat	1	Sahil
103	1002	Rajan	44444	4444444	saving	2	Atharva
	1003	Revati	555444				

updation anamoly→ if a customer tries to change the mobile number then it may get change only at one place, and will not get changed in all account

insertion anamoly--→if the customer donot open the account, then we will not be able to store the data of probable customers

deletion anamoly--→If any customer has only one account in the bank, and if customer closes the account, then along with account details, we will lose customer details also

account

accid	custid	balance	type	RM no
100	1001	455666	saving	1
101	1001	55555	current	1
102	1001	55666	demat	1
103	1002	4444444	saving	2

customer

custid	cname	mobile
1001	Kishori	7777
1002	Rajan	44444
1003	Revati	555444

RM

RM no	RM name
1	Sahil
2	Atharva

update customer

set mobile=6666

where custid=1001;

old

1001	Kishori	33333
------	---------	-------

new

1001	Kishori	6666
------	---------	------

create trigger before update on customer

for each row

update account

set mobile=new.mobile
where custid=old.custid

To normalize data

1. the data can be in 1NF,2NF,3NF,BCNF

1NF--- 1 normal form

1. all the values in the table should be atomic
2. each row and each column should have single value

Studid	sname	marks	subject
100	Rohit	78,88,99	java,c++,DBMS
101	amit	77,87,97	java,c++,DBMS

since in marks and subject column ,multiple values are stored, hence the table is not in 1NF

Studid	marks	subject
100	78	java
100	88	c++
100	99	DBMS
101	78	java
101	88	c++
101	99	DBMS

Student

Studid	sname
100	Rohit
101	amit

since in both tables each single row, single column contains single value so both of them are in 1NF

3. second normal form(2NF)

The table is in 2NF, if it is in 1NF

and no partial functional dependency should be there.

prime attribute---- all attributes which are part of candidate key

non prime attribute-----all attribute which are not part of candidate key are called as non prime attributes

studid	courseid	sname	cname	marks
123	100	Rohit	JAVA	89
123	101	Rohit	C++	89
124	100	Ramesh	JAVA	85
124	101	Ramesh	C++	87

It is in 1NF

to check whether it is 2NF

prime attributes--→ studid, courseid

nonprime attributes-→sname,cname,marks

find functional dependency.

studid--→sname

courseid $\rightarrow$ cname
studid+courseid $\rightarrow$ marks

if any non prime attribute is dependent on portion of the primary key, then partial functional dependency is there, and hence the table is not in 2NF

studid	sname
123	Rohit
124	Ramesh

courseid	cname
100	JAVA
101	C++

studid	courseid	marks
123	100	89
123	101	89
124	100	85
124	101	87

4. Third normal form(3NF)

if the table is in 2NF

if there is no transitive dependency is there, then it is in 3NF

$a \rightarrow b \rightarrow c$ then $a \rightarrow c$

prime attribute $\rightarrow$ non prime attribute $\rightarrow$ non prime attribute

accid	custid	cname	mobile	balance	type
100	1001	Kishori	33333	455666	saving
101	1001	Kishori	33333	55555	current
102	1001	Kishori	33333	55666	demat
103	1002	Rajan	44444	4444444	saving

accid $\rightarrow$ custid $\rightarrow$ cname

accid $\rightarrow$ custid $\rightarrow$ mobile

custid	cname	mobile
1001	Kishori	33333
1002	Rajan	44444

accid	custid	balance	type
100	1001	455666	saving
101	1001	55555	current
102	1001	55666	demat
103	1002	4444444	saving

BCNF (boyce codd normal form)

Boyce-Codd Normal Form (BCNF) is the advance version of the third normal form (3NF) that's why it is also known as a **3.5NF**

According to the E.F. Codd, a relation is in **Boyce-Codd normal form (3NF)** if it satisfies the following conditions:

- A relation is in 3NF.
- And, for every functional dependency, $X \rightarrow Y$, L.H.S of the functional dependency (X) be the super key of the table.

Example 3:

In this example, we have a relation R with three columns: studentId, Subject, and Professor. We have to find the highest normalization form, and also, if it is not in BCNF, we have to decompose it to satisfy the conditions of BCNF.

studentId	Subject	Professor
101	Java	Mayank
101	C++	Kartik
102	Java	Sarthak
103	C#	Lakshay
104	Java	Mayank

Interpreting the table:

- One student can enroll in more than one subject.
 - Example: student with **Id 101** has enrolled in **Java and C++**.
- Professor is assigned to the student for a specified subject, and there is always a possibility that there can be multiple professors teaching a particular subject.

Finding the solution:

- Using Id and Subject together, we can find all unique records and also the other columns of the table. Hence, the Id and Subject together form the primary key.
- The table is in 1NF because all the values inside a column are atomic and of the same domain.
- We can't uniquely identify a record solely with the help of either the Id or the Subject name. As there is no partial dependency, the table is also in 2NF.
- There is no [transitive dependency](#) because the non-prime attribute i.e., Professor, is not deriving any other non-prime attribute column in the table. Hence, the table is also in 3NF.
- There is a point to be noted that the table is not in **BCNF (Boyce-Codd Normal Form)**.

Why is the table not in BCNF?

As we know that each professor teaches only one subject, but one subject may be taught by multiple professors. This shows that there is a dependency between the subject & the professor, and the subject is always dependent on the professor (**professor -> subject**). As we know that the professor column is a non-prime attribute, while the subject is a prime attribute. This is not allowed in BCNF in DBMS. **For BCNF, the deriving attribute (professor here) must be a prime attribute.**

How to satisfy BCNF?

In Example 3, we will decompose the table into two tables: the Student table and the Professor table to satisfy the conditions of BCNF.

Student Table

	P_Id	S_Id	Professor
1	101	Mayank	
2	101	Kartik	
3	102	Sarthak	
4	103	Lakshay	
5	104	Mayank	

Professor Table

Professor	Subject
Mayank	Java
Kartik	C++
Sarthak	Java
Lakshay	C#

Professor is now the primary key and the prime attribute column, deriving the subject column. **Hence, it is in BCNF.**

In ER diagram 3 types of relations can be there

one-one	If the relation is one-one then any one side key can be added in other
one-Many	if the relation is one to many, then add key of one side to many side table
Many-Many	if relation is many to many then create 3 rd table and add primary key of both sides, and add all relation attributes

Proj	Proj	Proj	Empno	Ename	Grade	Sal	Proj	Alloc
Code	Type	Desc				scale	Join Date	Time

001	APP	LNG	46	JONES	A1	5	12/1/1998	24
001	APP	LNG	92	SMITH	A2	4	2/1/1999	24
001	APP	LNG	96	BLACK	B1	9	2/1/1999	18
004	MAI	SHO	72	JACK	A2	4	2/4/1999	6
004	MAI	SHO	92	SMITH	A2	4	5/5/1999	6

Table is in 1 NF

is it in 2NF

project code---→project type, project description,

empno-----→ename,grade, sal scale

project code+empno--→ project join date, alloc time

since partial functional dependency is there so the table is not in 2NF

Proj Code	Proj Type	Proj Desc	Empno	Ename	Grade	Sal scale	Proj Join Date	Alloc Time
001	APP	LNG	46	JONES	A1	5	12/1/1998 24	
001	APP	LNG	92	SMITH	A2	4	2/1/1999	24
001	APP	LNG	96	BLACK	B1	9	2/1/1999	18
004	MAI	SHO	72	JACK	A2	4	2/4/1999	6

Proj Code	Empno	Proj Join Date	Alloc Time
001	46	12/1/1998 24	
001	92	2/1/1999	24
001	96	2/1/1999	18
004	72	2/4/1999	6

project

Proj Code	Proj Type	Proj Desc
001	APP	LNG
004	MAI	SHO

Empno	Ename	Grade	Sal scale
46	JONES	A1	5
92	SMITH	A2	4
96	BLACK	B1	9
72	JACK	A2	4

empno-->grade->sal scale

proj_emp

Proj Code	Empno	Proj Join Date	Alloc Time
001	46	12/1/1998 24	
001	92	2/1/1999	24
001	96	2/1/1999	18
004	72	2/4/1999	6

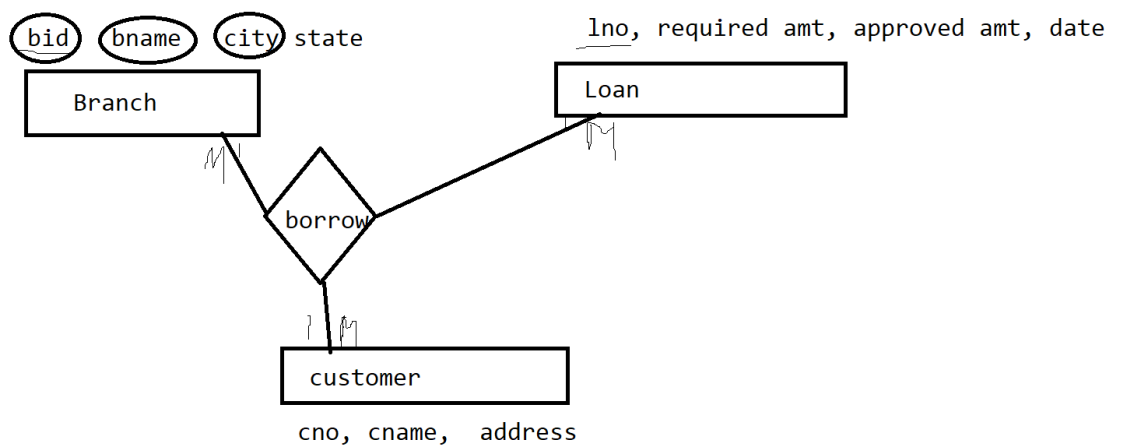
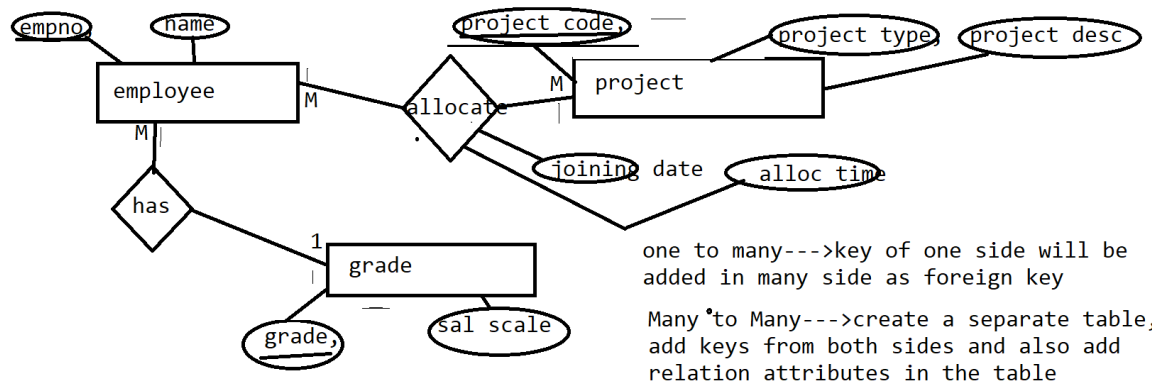
Is it in 3 NF

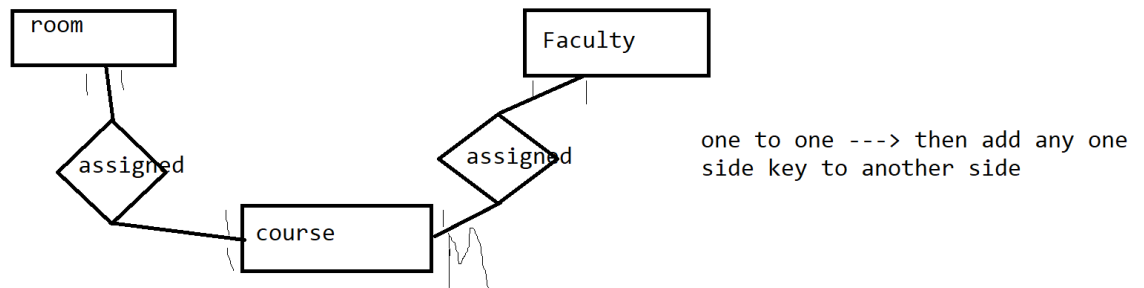
Employee

Empno	Ename	Grade
46	JONES	A1
92	SMITH	A2
96	BLACK	B1
72	JACK	A2

grade

Grade	Sal scale
A1	5
A2	4
B1	9





- Orderno
- Orderdate
- Itemno
- Qty
- Price
- Cname
- Custno
- Email
- Orderamt
- Salespersonno
- Salespersonname
- Locationid -----location from where item dispatched
- Location name

One customer can place many order

One order contains many items

One order will be managed by one salesperson

One order belong to one customer

One order can be dispatched from different location

stock item table --> item no, stock qty, stock price,date,

the given table is in 1NF

Is it in 2 NF ---->no

orderno--->orderdate,cname,cno,email,order amount,salesno,name

itemno

orderno+itemno->qty,price,locationid,lname

order

orderno, orderdate,cname,cno,email,order amount,salesno,sname

order-item

orderno,itemno,qty,price,locationid,lname

orderno->locationid->lname

These are no in 3NF, because we have transitive dependencies

orderno-> cno->cname, email

order no-→ salesno-→sname

order

orderno, orderdate,cno,order amount,salesno

customer

cno,cname,email

saleman

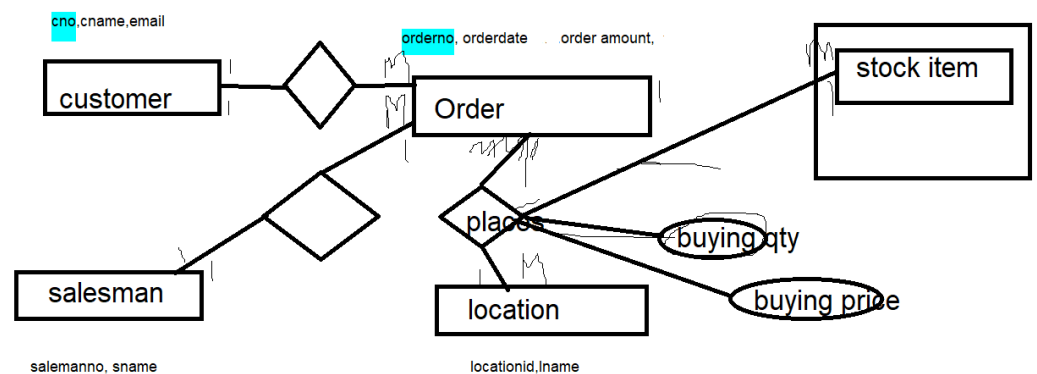
salemanno, sname

location

locationid,lname

order-item

orderno,itemno,qty,price,locationid



Movie ticket booking

movieid, mname, show time,duration, date,

price,capacity,seatno,screeno,screen_name, cname, cid, caddress, age, qty, mobile

screen no	seatno	date	show-time			
1	1		morning			
1	2					
1	3					

it is in 1 NF

Is it in 2NF

screenno--→capacity, screen_name

date

showtime

seatno

screenno+date

screen num+showtime

screen num+seatno

date+showtime

date+seatno

screennum+showtime+date-→movieid,mname,duration,price

screenum+showtime+date+seatno-→ cname, cid, caddress, age,qty,mobile

screen

screenno, capacity, screen_name

screen_movie

screennum+showtime+date, movieid, mname, duration, price

screen-cust

screennum+showtime+date+seatno, cname, cid, caddress, age, qty, mobile

to check it whether it is in 3 NF

screen_movie

screennum+showtime+date, movieid, mname, duration, price

screennum+showtime+date $\rightarrow$ movieid $\rightarrow$ mname, duration, price

screen_movie

screennum+showtime+date, movieid,

movie

movieid, mname, duration, price

screen-cust

screennum+showtime+date+seatno, cname, cid, caddress, age, qty, mobile

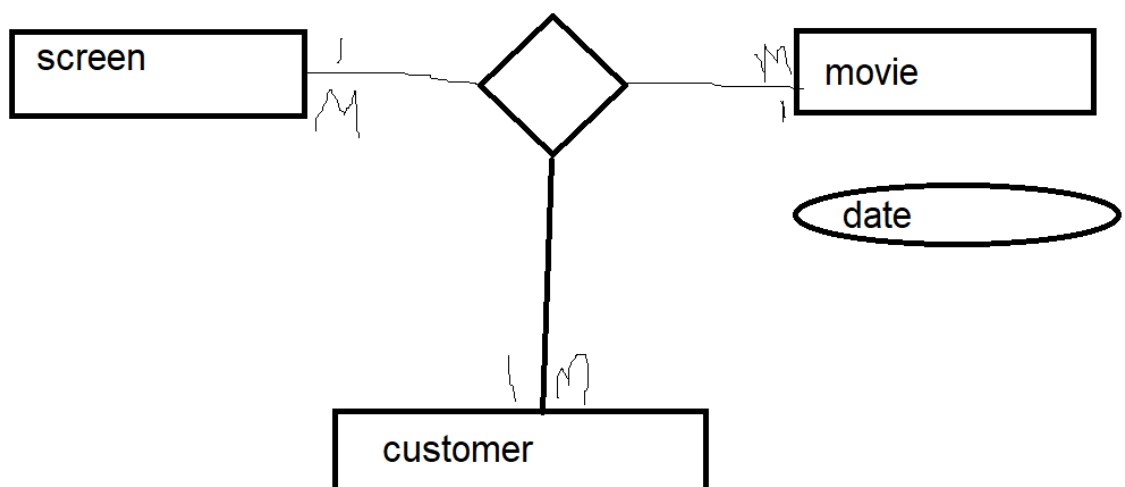
screennum+showtime+date+seatno $\rightarrow$ cid $\rightarrow$ cname, caddress, age, mobile

screen-cust

screennum+showtime+date+seatno, cid, qty

customer

cid, cname, caddress, age, mobile



- roomno
- customerno
- booking date
- roomname
- roomlocation
- cname
- address
- city
- state
- Email list
- phonenumber
- rateofroom
- mgrno
- managername

Rules

1. One customer can book many rooms
2. One room can be booked by many customers on different date
3. Booking of a customer is managed by one manager
4. One manager can book many room for many customer

